Better Than Guessing

Class 18 Instructor Notes | 15 minutes of a 75-minute period | Two-page handout | Laptops, teams of 2–3

Placement and purpose

Follows a long regression lecture that does the full SS_{Total} → fit → SS_{Residual} → R^2 calculation on the cricket data. The activity starts from that result rather than rebuilding it, and spends its time on the two things the lecture cannot do as well: where the line stops being trustworthy, and a case where R^2 points the wrong way.

The one idea: R^2 measures how much the line beat a flat guess at the mean — and it can be high for a model that is flatly wrong. The residual plot is what catches that.

No code blanks. Every cell runs as given; the skeleton is 17 cells. Part 1 is a single cell that reproduces the lecture's numbers, so students can check their lecture notes against it.

Timing

3 min	Part 1 — run the cricket fit, interpret the slope, square last week's r
4 min	Part 2 — extrapolation and the intercept
7 min	Part 3 — predict, then the twist and the residual plots
1 min	Buffer / closing line

If you are behind: take Q2.1 orally from the front — read the three predictions and ask which two to refuse. Do not cut Part 3; the residual plot is the only diagnostic in the activity that generalizes beyond this data set, and the predict-first in Q3.1 is what makes it land.

3 min — Part 1. The Cricket Line

quantity	value
mean temperature	80.04 °F
SS_{Total}	629.84
slope	3.2911
intercept	25.2323
SS_{Residual}	190.55
$R^2 = r^2$	0.6975
r	0.8351

Q1.1 slope in plain English: each additional chirp per second corresponds to about **3.3 °F** warmer. Push for the units — "degrees Fahrenheit per chirp per second" is the answer, and students who cannot state them usually cannot interpret the number either.

Q1.2 is the calibration question. Last class students found $r = 0.802$ between Pixar rating services and called it strong. Squared, that is **0.64** — barely two-thirds of the variation. Most students carry an intuition that $r = 0.8$ means "80% of the way to perfect." It does not. The crickets make the same point: $r = 0.84$ sounds excellent, $R^2 = 0.70$ sounds noticeably less so, and they are the same fit.

4 min — Part 2. Where the Line Stops Working

Predictions: **19 chirps → 87.8 °F** (fine, inside the observed range of 14.4–20.0); **40 chirps → 156.9 °F** (absurd — and no cricket chirps that fast); **0 chirps → 25.2 °F**, the intercept itself.

> **The intercept is the better half of this question.** Extrapolating to 40 is an obvious mistake that no one would make by accident. But the intercept is **printed in the regression output** and looks like a legitimate result — and it claims something about a temperature at which crickets have been silent for thirty degrees. Students who write "the intercept is the temperature when there are no chirps" have stated the arithmetic correctly and the science wrongly.

7 min — Part 3. A Better Number That Is a Worse Model

Second data set: $x = 0..6$, $y = x^2$. Fitted line $y = 6x - 5$, $SS_{\text{Total}} = 1092$, $SS_{\text{Residual}} = 84$, $\boldsymbol{R^2 = 0.9231}$.

Q3.1: expect many students, looking at a line that visibly misses a curve, to predict **lower**. It is higher (0.923 vs 0.698), and it is the one where a straight line is the **wrong** model. The crickets have a genuinely linear relationship with real scatter; the second set has no scatter at all and a systematic curve. Get a show of hands on the prediction before anyone runs the cell.

Q3.2 — the residuals. This is the payoff.

x	0	1	2	3	4	5	6
residual	+5	0	−3	−4	−3	0	+5

A perfect U. The cricket residuals, by contrast, scatter from −6.5 to +5.0 with no pattern — their correlation with chirp rate is exactly 0, which is true of **any** least-squares fit by construction, so what matters is the visible shape, not the number.

Q3.3: no. A high R^2 says the line is closer to the points than a flat mean is; it says nothing about whether a line was the right shape. The residual plot shows structure the line failed to capture — and here it shows it loudly, on the data set with the better R^2.

> **Closing line worth saying out loud:** every regression you fit from here on gets a residual plot before you report the R^2. It costs one cell.

Cut from the 40-minute version

The earlier version is in `_archive_40min/`. Three things came out:

- **Two baselines (paper).** Why the mean is the best single guess, and $R^2 = 1 / 0$ at the extremes. The lecture covers both; if it did not, ask the two limits orally in one minute.
- **The six-step walkthrough with code blanks.** The lecture does this calculation on the same data.
- **Negative $\boldsymbol{R^2}$.** Keeping the fitted slope and forcing $b = 0$ gives $R^2 \approx -14.5$. It needed a caveat — a true through-origin refit gives slope 4.79 and $R^2 = +0.55$ — and did not earn its time at 15 minutes. If a student asks whether R^2 can be negative: yes, for any prediction rule worse than the mean, but not for a least-squares line with an intercept.

The Cars 2 callback. Class 17′s notes promised that the diagnostic catching Cars 2 would arrive this class. Residual plots deliver only half of that: on the Pixar fit, Onward has the larger residual (−18.0) and Cars 2 the larger **leverage** (0.599 against a mean of 0.095). There is no room for it in this activity; if you want to close the loop, it belongs in the lecture. Otherwise say plainly that the second half of the answer is still coming.